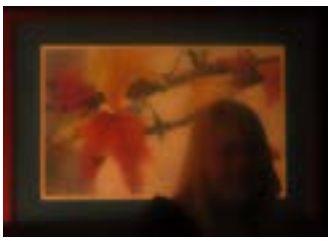
1. Bilateral filtering (100 points)

Implement bilateral filtering (40 points)
Implement joint-bilateral filtering (30 points)
Implement detail transfer (20 points)
Implement shadow and specularity masking (10 points)

In your report, compare the results of the four algorithms for different sets of parameter values. For these comparisons, make sure to show different images image1 − image2 , which should help highlight where in the images the techniques perform differently. Report the parameter values you chose as best for the four types of filtering, and discuss the advantages and disadvantages of each type.

Bilateral Filtering:

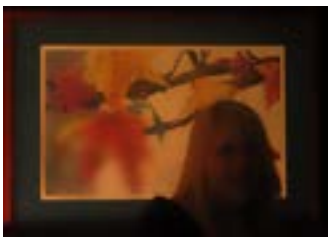
σs	**σr**	**Lamp Outcome**	**Museum Outcome**
-	-		
4	0.10		
16	0.05		
16	0.10		

16	0.20		
32	0.10		

The best parameters I found were σs=4 and σr = 0.10. The other values seem to either not give enough impact, or they drastically overblur the image, removing detail. Although σs=16 and σr = 0.05 also looked good, I found that the up close image of the σs=4 and σr = 0.10 parameters managed to keep the most relevant detail.

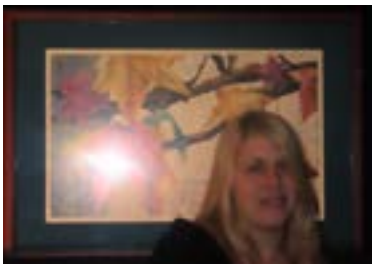
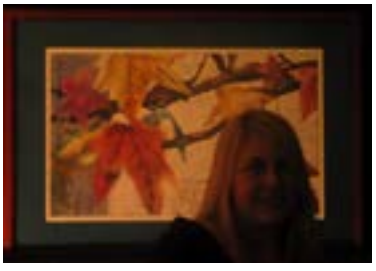
Joint-Bilateral Filtering:

σs	**σr**	**Lamp Outcome**	**Museum Outcome**
n/a	n/a		
4	0.10		
16	0.05		

16	0.10		
16	0.20		
32	0.10		

The best parameters I found were $\sigma s=4$ and $\sigma r = 0.10$. The other values seem to either not give enough impact, or they drastically overblur the image, removing detail.

Detail Transfer:

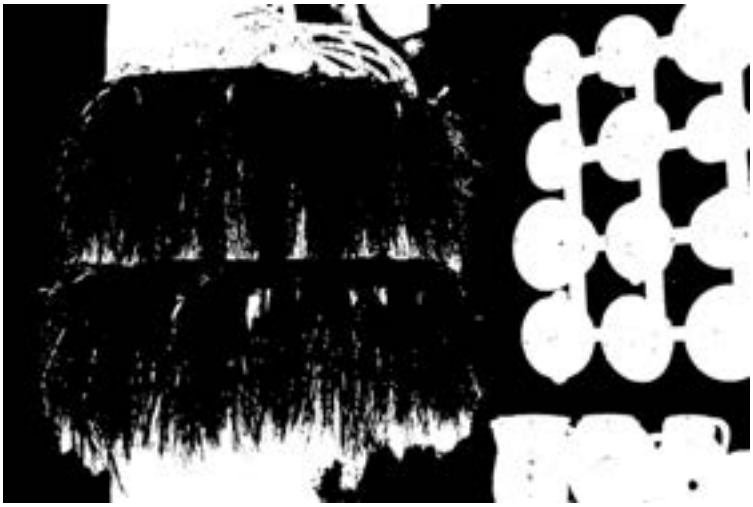
Parameters	**Lamp Outcome**	**Museum Outcome**
n/a		
0.001		

0.0001		

I selected epsilon=0.0001. The result is visually very similar to epsilon=0.001. A smaller epsilon allows stronger detail ratios in dark regions, while a larger epsilon makes those ratios more conservative.

Shadow and Specularity Masking:

White mask regions use the bilateral ambient estimate, and the black regions use detail transfer. I computed the mask from linearized luminance, scaling ambient by ISO_flash/ISO_ambient=200/1600=1/8 before comparison.





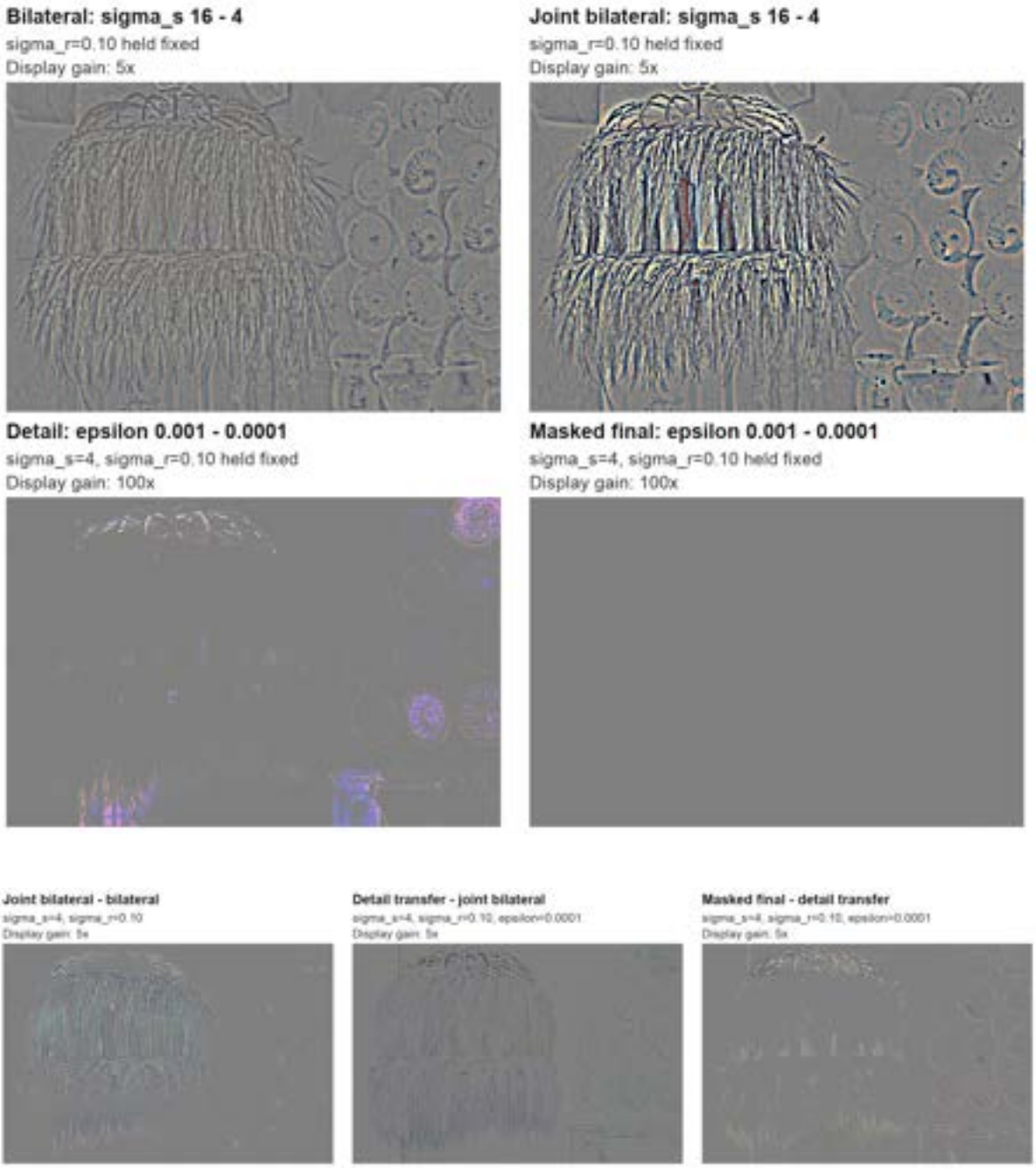
Overall:
Basic bilateral filtering reduces noise without requiring a flash image, but softens fine detail (like the detail that can be seen in the lamp). Joint bilateral filtering uses the cleaner flash edges as a guide, but cannot restore missing detail and can be affected by flash-only shadows or highlights. Detail transfer restores texture from the flash, but can also transfer those artifacts. Masked merging limits this transfer in unreliable regions by using the basic bilateral result there, at the cost of softer detail in masked areas.

Type	**Best Outcome**	**Parameters**
Bilateral Filtering		$\sigma s=4$ and $\sigma r = 0.10$
Joint-Bilateral Filtering		$\sigma s=4$ and $\sigma r = 0.10$

Detail Transfer		σs=4 and σr = 0.10 epsilon=0.0001
Shadow and Specularity Masking		σs=4 and σr = 0.10, epsilon=0.0001 shadow threshold=0.02 specularity threshold=0.95 morphology=3x3 mask feather sigma=2 pixels

Noise Before:	Noise in Final Result:
	

Lamp Difference:

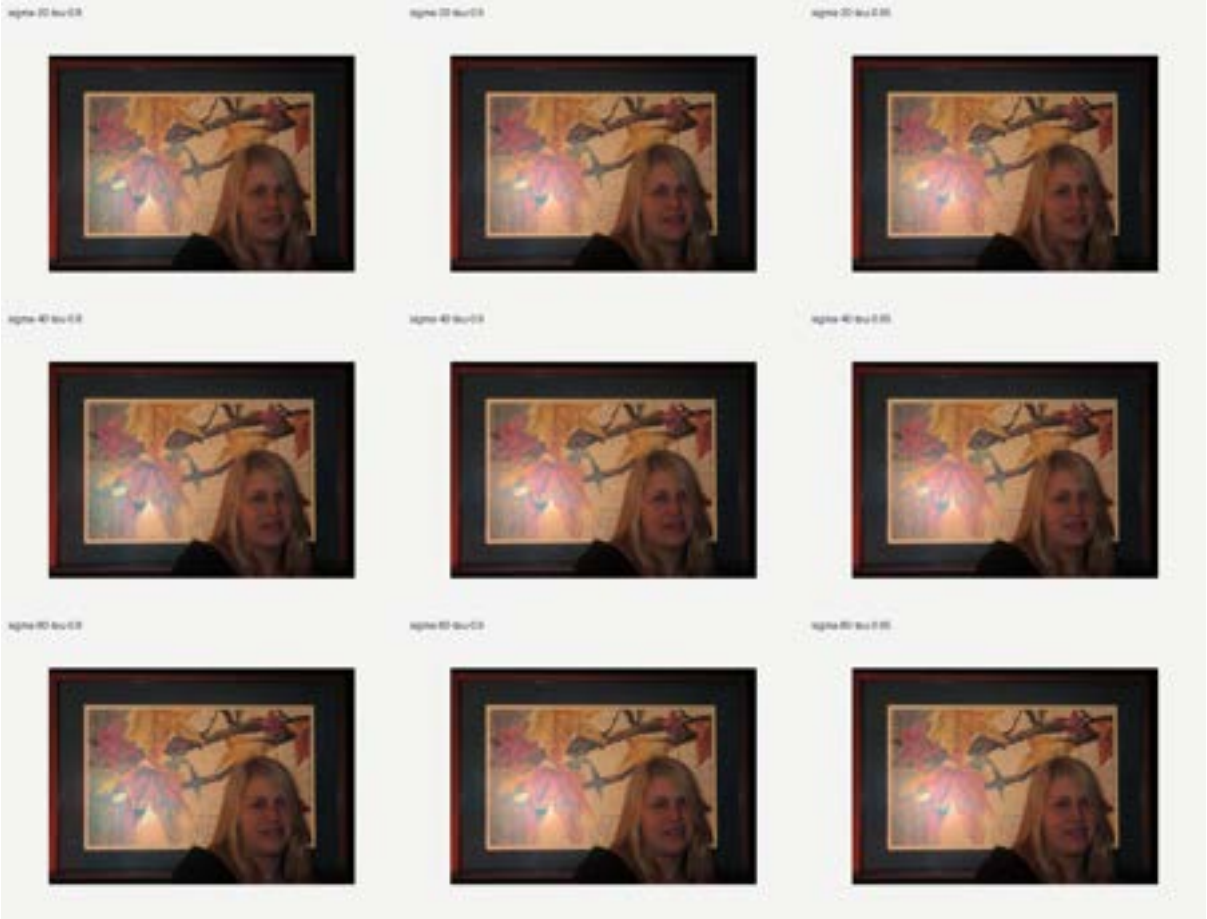


2. Gradient-domain processing (100 points)

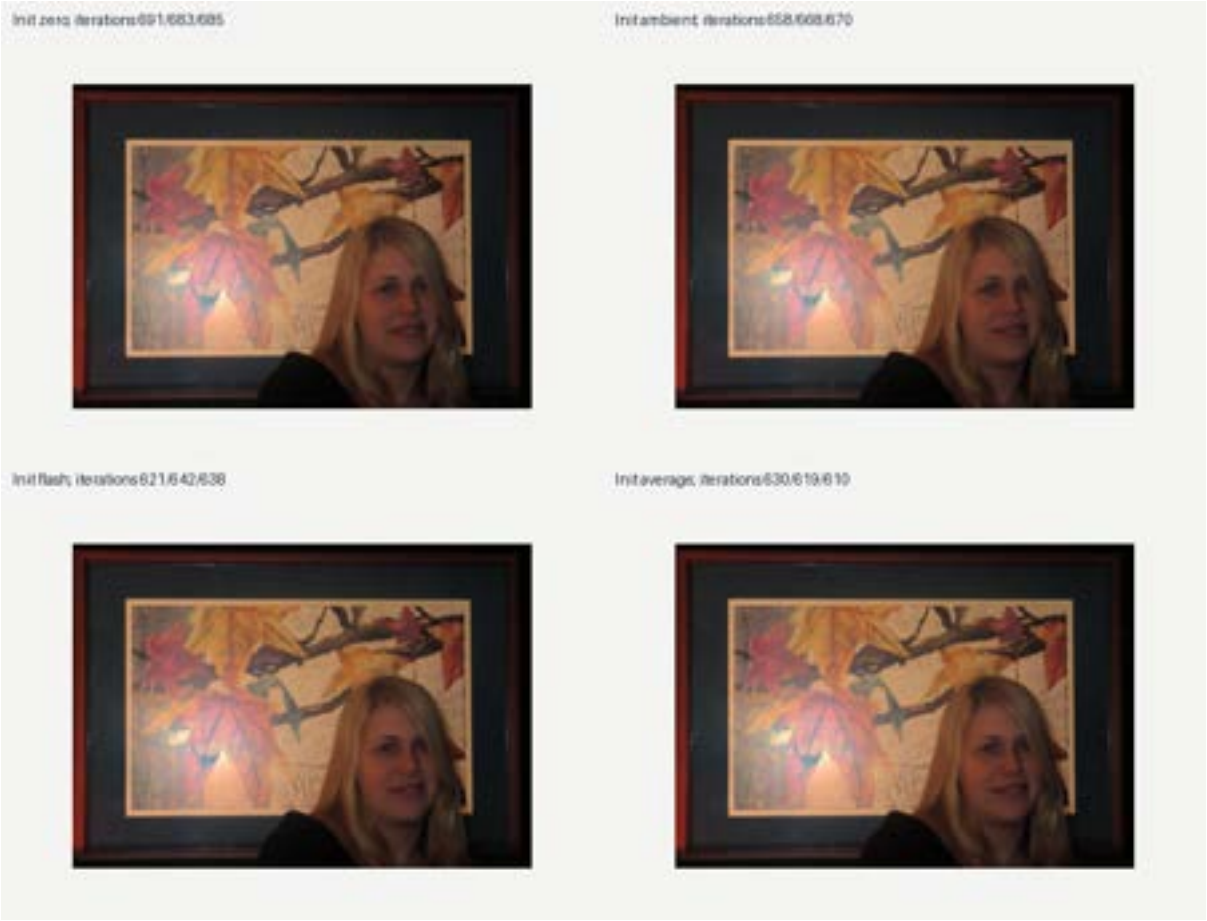
Differentiate and then re-integrate an image (75 points)
Create the fused gradient field (25 points)

Experiment with different values for the scalar parameters σ and τs, starting with the values recommended in the paper. Show the gradient fields ∇a, $\nabla\Phi'$, and $\nabla\Phi*$, and mention the parameter values you used.

I used forward differences for the gradient, backward differences for divergence, and the five-point Laplacian, verifying div(grad(I))=Laplacian(I) on interior pixels. I integrated the museum ambient gradient with Algorithm 1, starting from zero and fixing the outermost boundary to the original image. With epsilon=1e-6 and N=8000, reconstruction RMSE was 4.39e-7 and maximum absolute error was 1.73e-6 on the [0,1] scale; boundary pixels were unchanged.



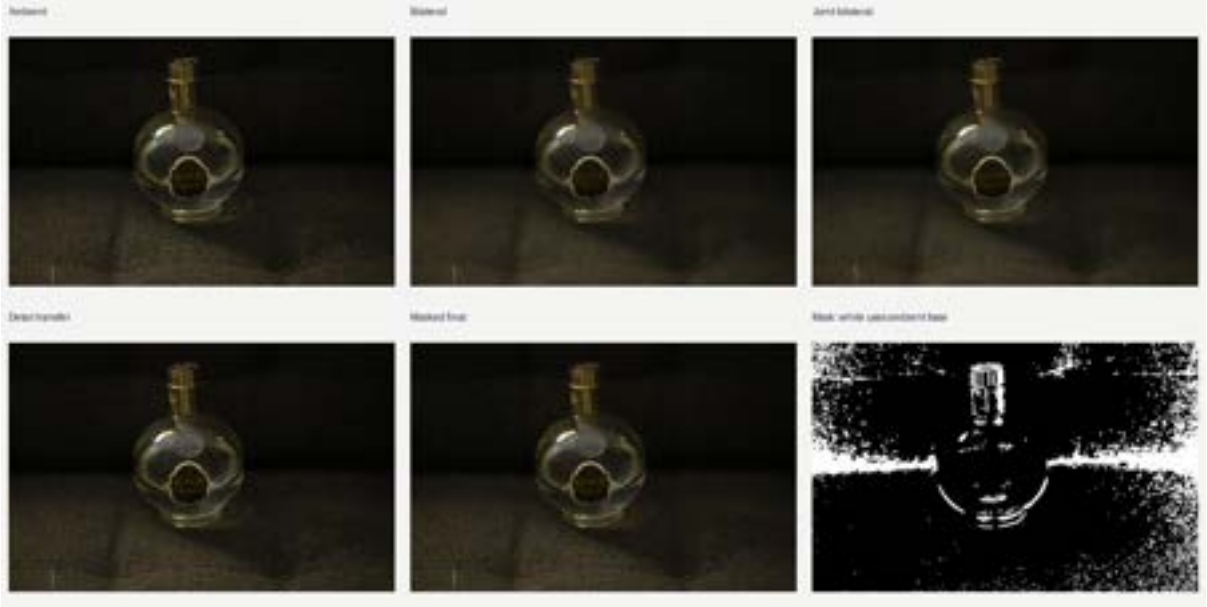
Experiment with different boundary conditions and initializations, and show the final fused image you obtain.





3. Capture your own flash/no-flash pairs (100 points)

The dim bottle scene is my bilateral-denoising example. After aligning the pair and cropping to their common valid region, I used sigma_s=8, sigma_r=0.10, epsilon=0.001, and shadow threshold=0.003. This reduces noise while preserving the warm ambient appearance, though some fine detail is smoothed. The glossy lemon dish is my gradient-fusion example: flash reveals the dim scene but introduces a strong highlight. I used sigma=40, tau_s=0.9, average boundaries, and average initialization. The result retains more ambient color while improving visibility, but does not eliminate all specularity. The cat is an additional gradient-fusion example using the same settings.

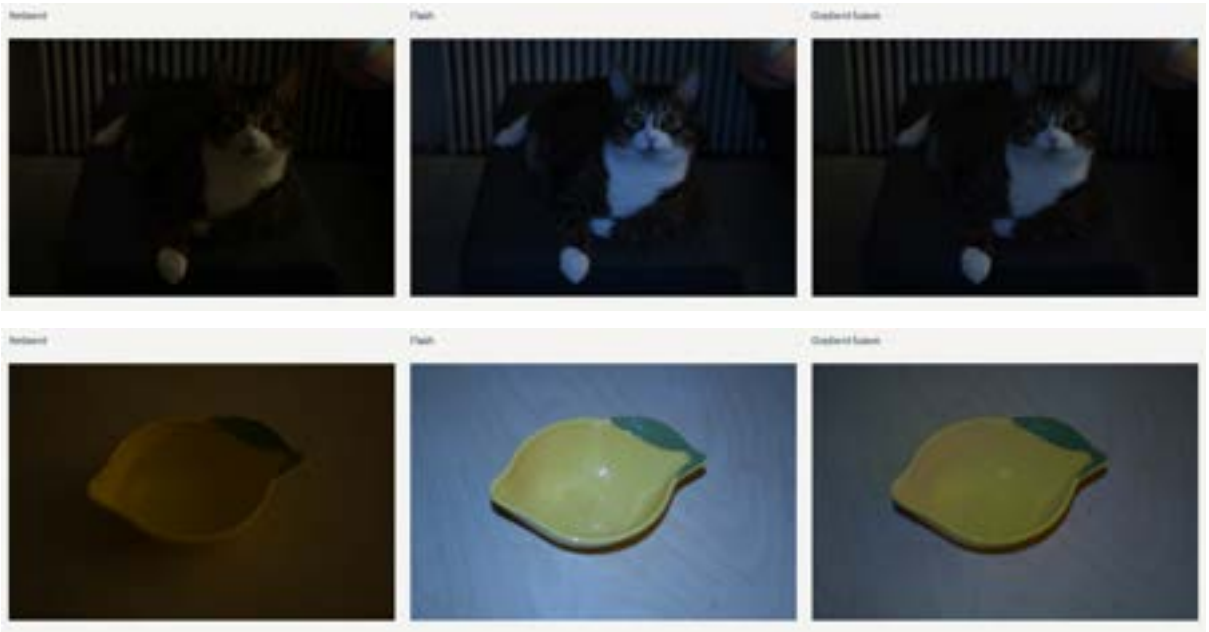




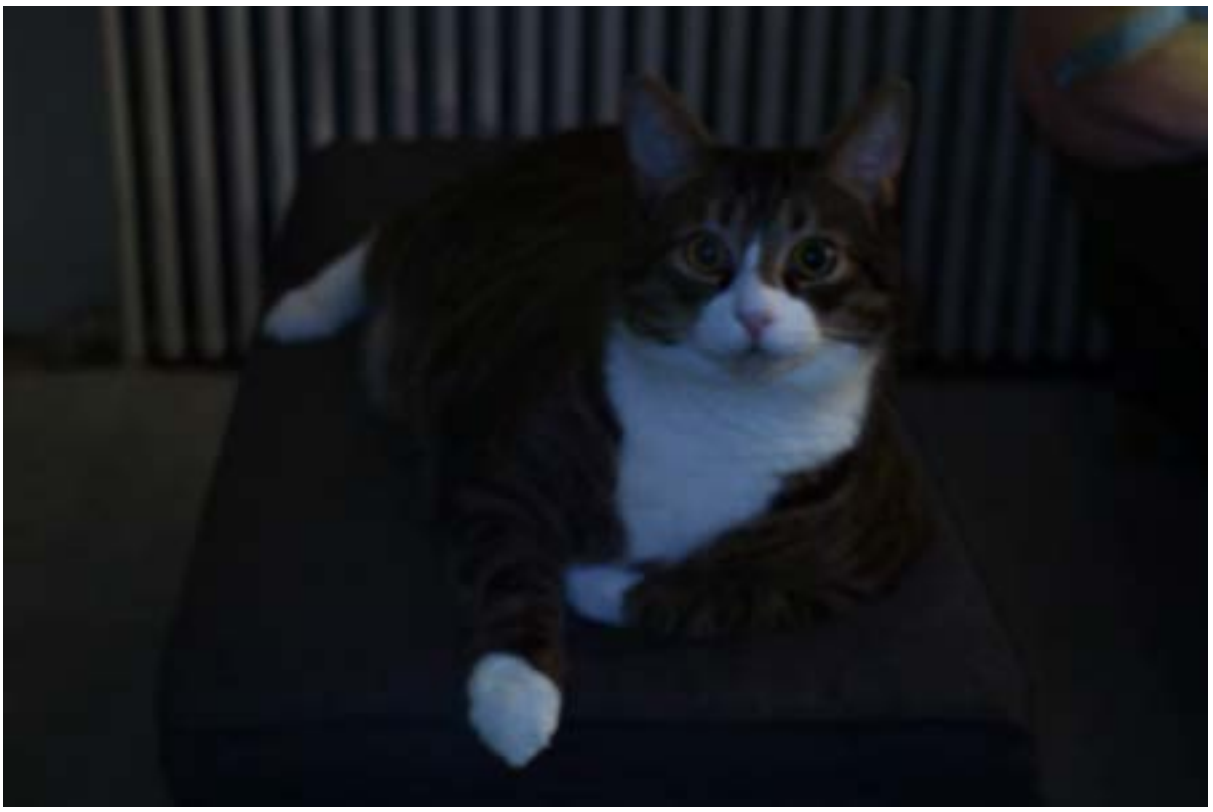
Final denoised image:



I still have no idea how I got my cat to sit still for two photos:



The writeup mentioned a photography competition, but I'm not sure if that's happening this semester. If there is one, this is my submission:



4. Bonus: Reflection removal in flash/no-flash photography (50 points)

5. Bonus: Other types of gradient-domain processing (up to 50 points)

I implemented a compact adaptation of GradientShop's NPR filter. It suppresses gradients in less salient regions and strengthens gradients near long contours, giving smoother color regions and stronger outlines. I kept the paper's edge-length gain, direction-dependent gradient scaling, and gradient weights. A conjugate-gradient solution minimizes the weighted gradient error plus a value multiplied by the difference between the original and NPR image, which keeps the colors and brightness close to the input. The original outer boundary is fixed. A light dark-edge overlay gives the final outlined appearance.

Images shown on next page:

potion_flash: input



potion_flash: NPR



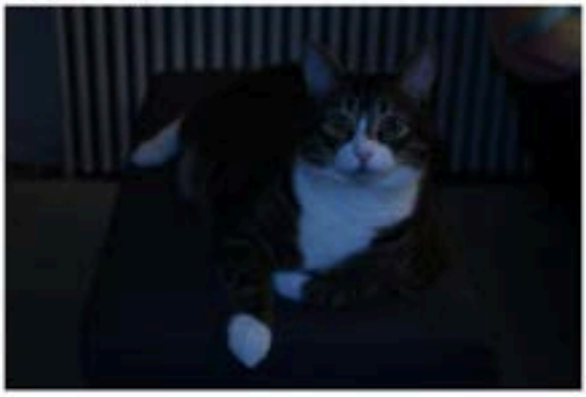
lemon_flash: input



lemon_flash: NPR



cat_flash: input



cat_flash: NPR

